

Output:-

Name: Uttam
Course: BCA
College: Coer University
Semester: 5th

Output:-

Age: 20
Marks: 85.5
Name: Uttam
Is Student: true

LAB SHEET - 01

- 1) Write a Dart program to display your name, course, college name, and semester.

```
void main () {  
  print ("Name: Uttam");  
  print ("Course: B.CA");  
  print ("College: Coer University");  
  print ("Semester: 5th");  
}
```

- 2) Write a dart program to declare variables of different data types (int, double, String, bool) and display their values.

```
void main () {  
  int age = 20;  
  double marks = 85.5;  
  String name = "Uttam";  
  bool isStudent = true;  
  print ("Age: $age");  
  print ("Marks: $marks");  
  print ("Name: $name");  
  print ("Is Student: $isStudent");  
}
```

Output:-

City: Roorkee
Country: India
Pi: 3.14159

Output:-

Sum: 30
Difference: 10
Product: 200
Quotient: 2.0

3) Write a dart program to demonstrate the use of var, final, and const.

```
void main() {  
    var city = "Roorkee";  
    final country = "India";  
    const pi = 3.14159;  
    print("City: $city");  
    print("Country: $country");  
    print("Pi: $pi");  
}
```

4) Write a dart program to accept two numbers and display their sum, difference, product, and quotient.

```
void main() {  
    int a = 20, b = 10;  
    print("Sum: ${a+b}");  
    print("Difference: ${a-b}");  
    print("Product: ${a*b}");  
    print("Quotient: ${a/b}");  
}
```

Output:-

Area : 15.0

Perimeter: 16.0

Output:-

Area of Circle : 153.93791

5) Write a Dart program to calculate the area and perimeter of a rectangle.

```
void main() {  
  double length = 5, breadth = 3;  
  double area = length * breadth;  
  double perimeter = 2 * (length + breadth);  
  print("Area: $area");  
  print("Perimeter: $perimeter");  
}
```

6) Write a Dart program to calculate the area of circle.

```
void main() {  
  double radius = 7;  
  double area = 3.14159 * radius * radius;  
  print("Area of circle: $area");  
}
```

Output:-

$$25.0^{\circ}\text{C} = 77.0^{\circ}\text{F}$$

Output:-

Before Swap: $a = 5, b = 10$

After Swap: $a = 10, b = 5$

7) WAD program to convert temperature from Celsius to Fahrenheit.

```
void main () {  
    double celsius = 25;  
    double fahrenheit = (celsius * 9/5) + 32;  
    print (" $celsius °C = $fahrenheit °F");  
}
```

8) WAD program to swap the values of two variables.

```
void main () {  
    int a = 5, b = 10;  
    print (" Before Swap: a = $a, b = $b");  
    int temp = a;  
    a = b;  
    b = temp;  
    print (" After Swap: a = $a, b = $b");  
}
```

Output:-
Negative

9) WAD program to check whether a given no. is positive, negative, or zero.

```
void main() {  
    int num = -3;  
    if (num > 0) {  
        print("Positive");  
    } else if (num < 0) {  
        print("Negative");  
    } else {  
        print("Zero");  
    }  
}
```

Output:-
odd

10) WAD program to check whether a given number is even or odd.

```
void main() {  
    int num = 7;  
    if (num % 2 == 0) {  
        print("Even");  
    } else {  
        print("odd");  
    }  
}
```

Output:-

Largest: 25

Output:-

Largest: 20

11) WAD program to find the largest of two numbers.

```
void main () {  
    int a = 15, b = 25;  
    printf ("Largest: ${a > b ? a : b}");  
}
```

12) WAD program to find the largest of three numbers.

```
void main () {  
    int a = 10, b = 20, c = 15;  
    int largest = (a > b) ? (a > c ? a : c) : (b > c ? b : c);  
    printf ("Largest: %d", largest);  
}
```

Output:-

2024 is a leap year

Output:-

Grade : B

13) WAD program to check whether a given year is a leap year.

```
void main() {  
    int year = 2024;  
    if ((year % 4 == 0 && year % 100 != 0) ||  
        (year % 400 == 0)) {  
        printf("$year is a Leap Year");  
    }  
    else {  
        printf("$year is not a Leap Year");  
    }  
}
```

14) WAD program to calculate grades based on the percentage obtained by a student.

```
void main() {  
    double percentage = 72;  
    if (percentage >= 90) {  
        printf("Grade: A+");  
    }  
    else if (percentage >= 75) {  
        printf("Grade: A");  
    }  
    else if (percentage >= 60) {  
        printf("Grade: B");  
    }  
    else if (percentage >= 40) {  
        printf("Grade: C");  
    }  
    else {  
        printf("fail");  
    }  
}
```

Output:-

1
2
3
4
5
6
7
8
9
10

Output:-

2
4
6
...
48
50

15) WAD program to print numbers from 1 to 10 using for loop.

```
void main() {  
    for (int i = 1; i <= 10; i++) {  
        print(i);  
    }  
}
```

16) WAD program to print all even numbers between 1 to 50.

```
void main() {  
    for (int i = 2; i <= 50; i += 2) {  
        print(i);  
    }  
}
```

Output:-

Sum: 55

Output:-

5x1 = 5
5x2 = 10
5x3 = 15
...
5x10 = 50

17) WAD program to calculate the sum of the first 10 natural numbers.

```
void main() {  
    int sum = 0;  
    for (int i = 1; i <= 10; i++) {  
        sum += i;  
    }  
    print("Sum: $sum");  
}
```

18) WAD program to generate the multiplication table of a given number.

```
void main() {  
    int num = 5;  
    for (int i = 1; i <= 10; i++) {  
        print("$num x $i = $ {num * i}");  
    }  
}
```

Output:- Factorial of 5 = 120

Output:- Reversed Number: 54321

19) WAD program to calculate the factorial of a given number.

```
void main() {  
    int num = 5;  
    int fact = 1;  
    for (int i = 1; i <= num; i++) {  
        fact *= i;  
    }  
    print ("factorial of $num = $fact");  
}
```

20) WAD program to reverse a given number using a loop.

```
void main() {  
    int num = 12345;  
    int reversed = 0;  
    while (num != 0) {  
        int digit = num % 10;  
        reversed = reversed * 10 + digit;  
        num /= 10;  
    }  
    print ("Reversed Number: $reversed");  
}
```

Output:-

Sum: 30

Output:-

29 is Prime

21) WAD function that accepts two numbers and returns their sum.

```
int addNumbers (int a, int b) {  
    return a+b;  
}  
  
void main () {  
    print ("Sum: $ {addNumbers (10,20)}");  
}
```

22) WAD function to check whether a given number is prime or not.

```
bool isPrime (int num) {  
    if (num <= 1) return false;  
    for (int i = 2; i <= num / 2; i++) {  
        if (num % i == 0) return false;  
    }  
    return true;  
}  
  
void main () {  
    int number = 29;  
    if (isPrime (number)) {  
        print ("$number is Prime");  
    } else {  
        print ("$number is Not Prime");  
    }  
}
```

Output:-

Aditya
Rahul
Sneha
Priya
Aman

Output:-

Largest: 67
Smallest: 3

23) Create a list of five student names and display each name using a loop.

```
void main() {  
    List<String> students = ["Aditya", "Rahul",  
                             "Sneha", "Priya", "Aman"];  
    for (String name in students) {  
        print(name);  
    }  
}
```

24) Create a list of numbers and find the largest and smallest number from the list.

```
void main() {  
    List<int> numbers = [12, 45, 3, 67, 23];  
    int largest = numbers[0];  
    int smallest = numbers[0];  
    for (int num in numbers) {  
        if (num > largest) largest = num;  
        if (num < smallest) smallest = num;  
    }  
    print("Largest: $largest");  
    print("Smallest: $smallest");  
}
```

Output:-

Name: Uttam

Total Marks: 225

Percentage: 75.0

Result: Pass

25) Develop a Student Result Program in dart that stores the student's name and marks in three subjects, calculates total marks and percentage, and display the result as Pass or Fail.

```
void main () {  
  String name = "Uttam";  
  int sub1 = 75, sub2 = 82, sub3 = 68;  
  int total = sub1 + sub2 + sub3;  
  double percentage = total / 3;  
  print ("Name: $name");  
  print ("Total Marks: $total");  
  print ("Percentage: $percentage");  
  if (percentage >= 40) {  
    print ("Result: Pass");  
  } else {  
    print ("Result: Fail");  
  }  
}
```

[Signature]
24/8/26